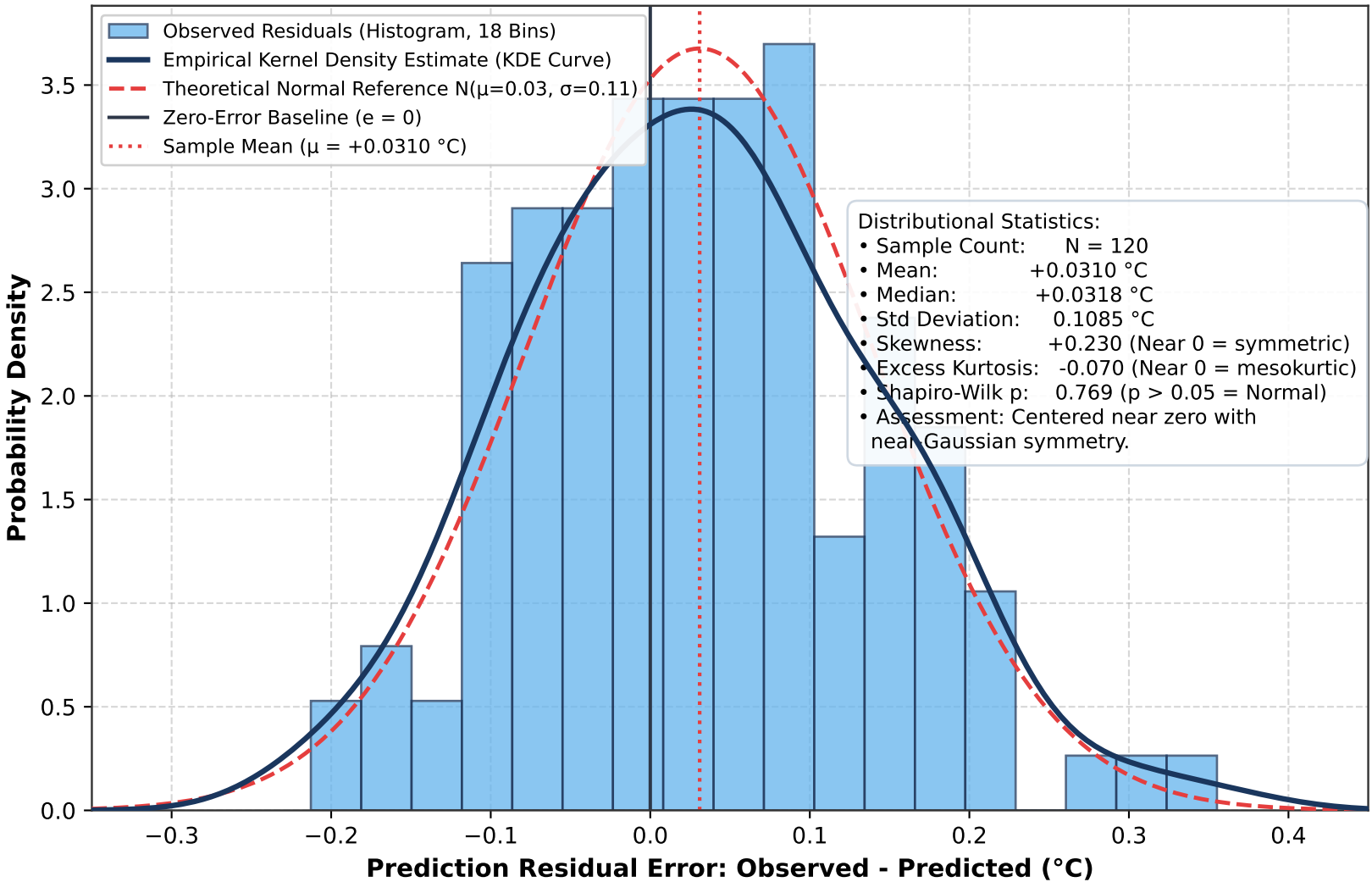
Graph 4: Residual Distribution and Density Analysis
Evaluation of Ridge Regression Out-of-Sample Residuals (2016–2025)
Observed Residuals (Histogram, 18 Bins)
Empirical Kernel Density Estimate (KDE Curve)
Theoretical Normal Reference N(μ=0.03, σ=0.11)
Zero-Error Baseline (e = 0)
Sample Mean (μ = +0.0310 °C)
Distributional Statistics:
• Sample Count: N = 120
• Mean: +0.0310 °C
• Median: +0.0318 °C
• Std Deviation: 0.1085 °C
• Skewness: +0.230 (Near 0 = symmetric)
• Excess Kurtosis: -0.070 (Near 0 = mesokurtic)
• Shapiro-Wilk p: 0.769 (p > 0.05 = Normal)
• Assessment: Centered near zero with near-Gaussian symmetry.
Probability Density
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
Prediction Residual Error: Observed - Predicted (°C)